

8 2025-01-30 | Week 03 | Lecture 08

This lecture is based on sections 2.3 and 2.4 in the textbook.

8.1 Applications of the 3 combinatorial principles

Example 26. Let $S = \{1, 2, 3, 4, 5, 6\}$. How many unordered subsets (i.e., combinations) of size 2 are there?

$$\binom{6}{2} = \frac{6!}{2! \cdot 4!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(2 \cdot 1) \cdot (4 \cdot 3 \cdot 2 \cdot 1)} = \frac{6 \cdot 5}{2} = 15.$$

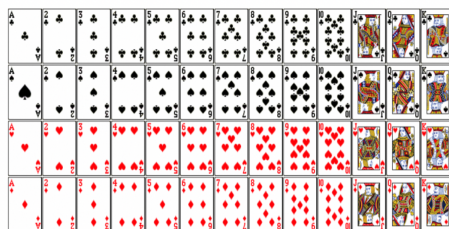
The fifteen combinations are

$$\begin{aligned} &\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{1, 6\} \\ &\{2, 3\}, \{2, 4\}, \{2, 5\}, \{2, 6\} \\ &\{3, 4\}, \{3, 5\}, \{3, 6\} \\ &\{4, 5\}, \{4, 6\} \\ &\{5, 6\} \end{aligned}$$

note that the order doesn't matter, i.e., $\{2, 1\} = \{1, 2\}$ so this just counts as one, not two.

End of Example 26. $\square$

Example 27 (Poker). A poker hand consist of 5 randomly chosen cards from a standard 52-card deck.



Some questions about poker hands:

1. How many distinct poker hands are there?

$$\binom{52}{5} = \frac{52!}{5! \cdot 47!} = 2,598,960$$

2. A **flush** occurs when all five cards are the same suit. How many ways are there to get a flush?

$$4 \times \binom{13}{5} = 4 \cdot \frac{13!}{5! 8!} = 4 \cdot \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 5148$$

The 4 is to choose a suit. Then we choose 5 out of 13 cards in that suit.

3. What's the probability of getting a flush?

To answer this, we use the enumeration principle:

$$\frac{\binom{4}{1} \times \binom{13}{5}}{\binom{52}{5}} = \frac{5,148}{2,598,960} \approx 0.002$$

In other words, the probability of a flush is pretty low: about 0.2%.

End of Example 27. $\square$

8.2 Introduction to craps

Craps is a dice-rolling game in which a player, known as the shooter, rolls a pair of 6-sided dice. The shooter wins, loses or re-rolls depending on what happens.

Here's a set of rules for the shooter:

1. You roll two dice and add them up.
 - If the sum is 2, 3, or 12, you lose (“craps”)
 - If you roll 7 or 11, then you win (“natural”)
 - Otherwise you establish “point”, which is whatever number you got.
2. If you established point, then you must keep rolling until one of two events occurs:
 - You roll a 7: you lose
 - You roll your “point” number: you win.

Tallying the results, we saw that shooters won about 50% of the time. Is this a fair game? We will set out to answer this question. To do so, we'll need to introduce some new ideas. A key idea is that of conditional probability:

8.3 Conditional probability

This section is based on section 2.4 in the text.

Definition 28 (Conditional Probability). Let A, B be events, and assume that $\mathbb{P}[A] > 0$. Then the **conditional probability of B , given A** , denoted $\mathbb{P}[B \mid A]$, is given by the formula

$$\mathbb{P}[B \mid A] := \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[A]}.$$